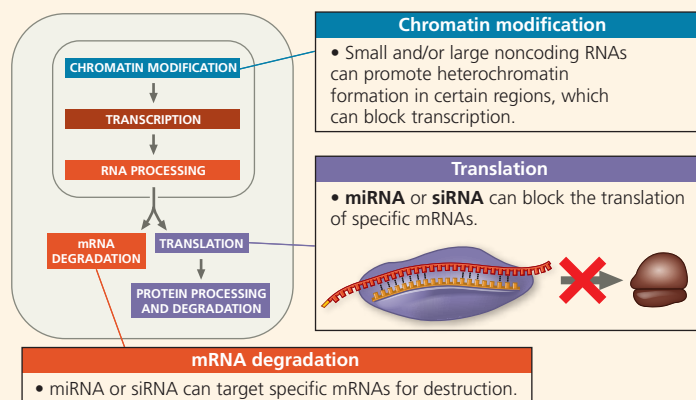


CONCEPT 18.3

Noncoding RNAs play multiple roles in controlling gene expression (pp. 379–381)



? Why are miRNAs called noncoding RNAs? Explain how they participate in gene regulation.

CONCEPT 18.4

A program of differential gene expression leads to the different cell types in a multicellular organism (pp. 381–388)

- Embryonic cells become committed to a certain fate (**determination**), and undergo **differentiation**, becoming specialized in structure and function for their determined fate. Cells differ in structure and function not because they contain different genomes but because they express different genes. **Morphogenesis** encompasses the processes that give shape to the organism and its various structures.
- Localized **cytoplasmic determinants** in the unfertilized egg are distributed differentially to daughter cells, where they regulate the expression of those cells' developmental fates. In the process called **induction**, signaling molecules from embryonic cells cause transcriptional changes in nearby target cells.
- Differentiation is marked by the appearance of tissue-specific proteins, which enable differentiated cells to carry out their specialized roles.
- In animals, **pattern formation**, the development of a spatial organization of tissues and organs, begins in the early embryo. **Positional information**, the molecular cues that control pattern formation, tells a cell its location relative to the body's axes and to other cells. In *Drosophila*, gradients of **morphogens** encoded by **maternal effect genes** determine the body axes. For example, the gradient of Bicoid protein determines the anterior-posterior axis.

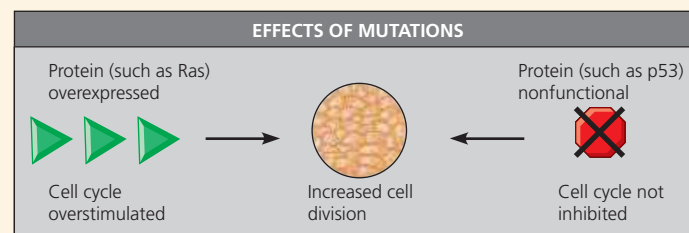
? Describe the two main processes that cause embryonic cells to head down different pathways to their final fates.

CONCEPT 18.5

Cancer results from genetic changes that affect cell cycle control (pp. 388–395)

- The products of **proto-oncogenes** and **tumor-suppressor genes** control cell division. A DNA change that makes a proto-oncogene overly active converts it to an **oncogene**, which may promote extra cell division and cancer. A tumor-suppressor gene encodes a protein that inhibits abnormal cell division. A mutation that reduces the activity of its protein product may lead to excessive cell division and cancer.

- Many proto-oncogenes and tumor-suppressor genes encode components of growth-stimulating and growth-inhibiting signaling pathways, respectively, and mutations in them can interfere with normal cell-signaling pathways. A hyperactive version of a protein in a stimulatory pathway, such as **Ras** (a G protein), functions as an oncogene protein. A defective version of a protein in an inhibitory pathway, such as **p53** (a transcription activator), fails to act as a tumor suppressor.



- In the multistep model of cancer development, normal cells are converted to cancer cells by the accumulation of mutations affecting proto-oncogenes and tumor-suppressor genes. Technical advances in DNA and mRNA sequencing are enabling cancer treatments that are more individually based.
- Genomics-based studies have resulted in researchers proposing four subtypes of breast cancer, based on expression of genes by tumor cells.
- Individuals who inherit a mutant allele of a proto-oncogene or tumor-suppressor gene have a predisposition to develop a particular cancer. Certain viruses promote cancer by integration of viral DNA into a cell's genome.

? Compare the usual functions of proteins encoded by proto-oncogenes with those of proteins encoded by tumor-suppressor genes.

TEST YOUR UNDERSTANDING

➔ For more multiple-choice questions, go to the **Practice Test** in the **Mastering Biology** eText or Study Area, or go to goo.gl/GruWRg.

Levels 1-2: Remembering/Understanding

- If a particular operon encodes enzymes for making an essential amino acid and is regulated like the *trp* operon, then
(A) the amino acid inactivates the repressor.
(B) the repressor is active in the absence of the amino acid.
(C) the amino acid acts as a corepressor.
(D) the amino acid turns on transcription of the operon.
- Muscle cells differ from nerve cells mainly because they
(A) express different genes.
(B) contain different genes.
(C) use different genetic codes.
(D) have unique ribosomes.
- The functioning of enhancers is an example of
(A) a eukaryotic equivalent of prokaryotic promoter functioning.
(B) transcriptional control of gene expression.
(C) the stimulation of translation by initiation factors.
(D) post-translational control that activates certain proteins.
- Cell differentiation always involves
(A) transcription of the *myoD* gene.
(B) the movement of cells.
(C) the production of tissue-specific proteins.
(D) the selective loss of certain genes from the genome.
- Which of the following is an example of post-transcriptional control of gene expression?
(A) the addition of methyl groups to cytosine bases of DNA
(B) the binding of transcription factors to a promoter
(C) the removal of introns and alternative splicing of exons
(D) gene amplification contributing to cancer